

## PUKASKWA, ON, Canada

WMO: 717500

Lat: 48.608N

Lon: 86.287W

Elev: 681

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-22.1 | (c)<br>-16.0 | (d)<br>-28.5                  | (e)<br>1.1 | (f)<br>-21.8 | (g)<br>-22.5 | (h)<br>1.6 | (i)<br>-15.3 | (j)<br>36.0           | (k)<br>24.3 | (l)<br>31.9 | (m)<br>26.1 | (n)<br>6.3            | (o)<br>50 | (p)<br>0.702 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>14.0            | (c)<br>73.6     | (d)<br>63.6 | (e)<br>70.9 | (f)<br>63.3 | (g)<br>68.7 | (h)<br>62.3 | (i)<br>66.9         | (j)<br>70.5 | (k)<br>65.4 | (l)<br>68.7 | (m)<br>63.9 | (n)<br>67.1 | (o)<br>9.0           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.6                            | 96.9 | 68.5 | 64.0 | 91.6 | 67.0 | 62.5 | 86.8 | 65.7 | 31.9          | 70.3 | 30.6 | 68.9 | 29.5 | 67.1 | 73.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| Min               | Max  |      | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c)  |                            |     | (e)                | (f)  | (g)                                                | (h) | (i)        | (j)  | (k)        | (l)  | (m)        | (n)  | (o)   | (p)  |
| 28.5              | 24.7 | 20.9 | DB                         |     | -28.5              | 81.1 | 7.6                                                | 3.1 | -34.0      | 83.3 | -38.4      | 85.1 | -42.7      | 86.8 | -48.2 | 89.1 |
|                   |      |      | WB                         |     | -28.7              | 70.1 | 7.6                                                | 2.2 | -34.1      | 71.7 | -38.6      | 73.0 | -42.8      | 74.2 | -48.3 | 75.8 |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 36.8   | 9.1   | 13.1  | 23.0  | 34.8  | 45.7  | 53.5  | 58.0  | 60.1  | 54.1  | 42.6  | 29.7  | 16.9  |
|                                                                   | DBStd       | 19.87  | 14.11 | 12.76 | 11.75 | 8.05  | 7.10  | 5.81  | 5.53  | 5.05  | 7.10  | 7.68  | 10.47 | 13.13 |
|                                                                   | HDD50       | 5734   | 1268  | 1033  | 838   | 459   | 174   | 28    | 5     | 1     | 39    | 255   | 609   | 1025  |
|                                                                   | HDD65       | 10304  | 1733  | 1453  | 1302  | 907   | 600   | 346   | 226   | 164   | 331   | 694   | 1058  | 1490  |
|                                                                   | CDD50       | 933    | 0     | 0     | 1     | 2     | 40    | 134   | 252   | 314   | 163   | 26    | 1     | 0     |
|                                                                   | CDD65       | 27     | 0     | 0     | 0     | 0     | 0     | 2     | 8     | 13    | 4     | 0     | 0     | 0     |
|                                                                   | CDH74       | 83     | 0     | 0     | 0     | 0     | 4     | 20    | 30    | 23    | 6     | 0     | 0     | 0     |
|                                                                   | CDH80       | 9      | 0     | 0     | 0     | 0     | 0     | 3     | 2     | 3     | 1     | 0     | 0     | 0     |
|                                                                   | WSAvg       | 8.5    | 9.4   | 9.1   | 8.6   | 8.0   | 7.4   | 6.6   | 6.6   | 7.8   | 9.2   | 9.7   | 9.8   | 9.6   |
|                                                                   | PrecAvg     | 35.1   | 2.3   | 1.8   | 1.9   | 2.6   | 3.3   | 3.5   | 3.4   | 3.4   | 4.1   | 3.7   | 2.6   | 2.4   |
| Precipitation                                                     | PrecMax     | 42.0   | 5.0   | 3.5   | 3.3   | 5.3   | 4.8   | 6.4   | 5.9   | 5.5   | 6.8   | 5.5   | 4.6   | 4.0   |
|                                                                   | PrecMin     | 27.1   | 1.5   | 0.6   | 0.7   | 1.2   | 1.5   | 1.5   | 0.9   | 1.4   | 1.5   | 1.7   | 1.5   | 1.2   |
|                                                                   | PrecStd     | 3.8    | 0.7   | 0.7   | 0.6   | 0.9   | 0.8   | 1.1   | 1.2   | 1.1   | 1.3   | 1.2   | 0.7   | 0.7   |
|                                                                   | PrecStd     | 3.8    | 0.7   | 0.7   | 0.6   | 0.9   | 0.8   | 1.1   | 1.2   | 1.1   | 1.3   | 1.2   | 0.7   | 0.7   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 38.8   | 41.6  | 56.6  | 60.6  | 72.2  | 76.4  | 78.0  | 76.6  | 73.3  | 64.8  | 54.3  | 42.5  | 29.7  |
|                                                                   | 0.4% MCWB   | 38.0   | 38.2  | 49.4  | 49.1  | 57.2  | 61.7  | 64.4  | 64.4  | 65.2  | 58.5  | 50.3  | 40.7  | 29.7  |
|                                                                   | 2% DB       | 36.1   | 37.6  | 45.9  | 54.5  | 65.1  | 70.6  | 73.1  | 72.7  | 69.6  | 61.0  | 50.1  | 40.0  | 29.7  |
|                                                                   | 2% MCWB     | 35.0   | 35.7  | 42.1  | 44.9  | 56.3  | 59.6  | 63.8  | 65.0  | 63.7  | 56.8  | 47.5  | 38.5  | 29.7  |
|                                                                   | 5% DB       | 34.1   | 34.7  | 41.3  | 50.2  | 61.0  | 66.8  | 70.0  | 70.5  | 67.2  | 58.3  | 46.6  | 38.2  | 29.7  |
|                                                                   | 5% MCWB     | 33.0   | 33.2  | 37.9  | 42.2  | 53.2  | 58.6  | 62.7  | 64.1  | 62.8  | 54.8  | 44.3  | 36.7  | 29.7  |
|                                                                   | 10% DB      | 31.3   | 32.1  | 37.8  | 46.5  | 57.0  | 63.7  | 67.2  | 68.7  | 65.0  | 54.9  | 44.1  | 35.5  | 29.7  |
|                                                                   | 10% MCWB    | 30.1   | 30.6  | 34.9  | 40.2  | 49.9  | 56.9  | 61.7  | 63.1  | 61.4  | 51.9  | 42.1  | 33.9  | 29.7  |
|                                                                   | 0.4% WB     | 37.8   | 39.0  | 50.4  | 51.1  | 62.9  | 66.3  | 68.7  | 69.4  | 67.3  | 61.1  | 51.6  | 41.4  | 29.7  |
|                                                                   | 0.4% MCDB   | 38.2   | 40.5  | 54.2  | 55.7  | 65.1  | 71.3  | 72.8  | 72.7  | 70.6  | 62.9  | 52.8  | 41.9  | 29.7  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 35.3   | 35.7  | 42.7  | 46.8  | 58.3  | 63.1  | 66.1  | 67.3  | 65.5  | 57.4  | 47.5  | 38.9  | 29.7  |
|                                                                   | 2% MCDB     | 35.9   | 37.1  | 45.3  | 52.5  | 63.1  | 67.4  | 70.2  | 70.7  | 68.1  | 60.1  | 49.5  | 39.8  | 29.7  |
|                                                                   | 5% WB       | 33.1   | 33.4  | 38.4  | 43.5  | 54.4  | 60.5  | 64.3  | 65.7  | 63.7  | 54.9  | 44.8  | 36.7  | 29.7  |
|                                                                   | 5% MCDB     | 34.0   | 34.5  | 40.8  | 48.8  | 59.4  | 64.4  | 68.1  | 69.0  | 66.3  | 57.8  | 46.5  | 38.1  | 29.7  |
|                                                                   | 10% WB      | 30.0   | 30.8  | 35.4  | 40.6  | 50.7  | 58.1  | 62.5  | 64.2  | 61.8  | 52.4  | 42.4  | 34.2  | 29.7  |
|                                                                   | 10% MCDB    | 31.3   | 32.2  | 37.3  | 45.3  | 55.4  | 62.0  | 66.2  | 67.3  | 64.6  | 54.9  | 44.1  | 35.4  | 29.7  |
| Mean Daily Temperature Range                                      | MDBR        | 18.6   | 18.9  | 18.6  | 16.2  | 16.5  | 16.8  | 14.6  | 14.0  | 13.8  | 12.2  | 11.9  | 15.3  | 15.3  |
|                                                                   | 5% DB       | 13.0   | 13.7  | 16.3  | 20.9  | 22.0  | 21.7  | 18.6  | 15.7  | 14.2  | 14.9  | 12.5  | 12.9  | 12.9  |
|                                                                   | MCWBR       | 12.9   | 12.3  | 11.8  | 13.2  | 13.6  | 12.6  | 10.7  | 9.4   | 9.8   | 11.0  | 11.2  | 12.8  | 12.8  |
|                                                                   | 5% WB       | 13.2   | 13.5  | 14.8  | 18.1  | 19.7  | 18.5  | 15.4  | 13.0  | 12.6  | 13.4  | 12.2  | 12.9  | 12.9  |
|                                                                   | MCWBR       | 13.3   | 12.7  | 11.4  | 12.8  | 14.1  | 13.6  | 10.6  | 8.8   | 9.7   | 10.6  | 11.1  | 12.9  | 12.9  |
|                                                                   | MCWBR       | 13.3   | 12.7  | 11.4  | 12.8  | 14.1  | 13.6  | 10.6  | 8.8   | 9.7   | 10.6  | 11.1  | 12.9  | 12.9  |
| Clear-Sky Solar Irradiance                                        | taub        | 0.269  | 0.270 | 0.306 | 0.341 | 0.363 | 0.370 | 0.387 | 0.369 | 0.360 | 0.325 | 0.289 | 0.262 | 0.262 |
|                                                                   | taud        | 2.396  | 2.407 | 2.366 | 2.357 | 2.402 | 2.404 | 2.394 | 2.454 | 2.463 | 2.529 | 2.528 | 2.454 | 2.454 |
|                                                                   | Ebn at Noon | 257    | 285   | 289   | 288   | 285   | 283   | 276   | 277   | 268   | 261   | 250   | 246   | 246   |
|                                                                   | Edh at Noon | 21     | 26    | 32    | 36    | 36    | 36    | 36    | 32    | 29    | 23    | 18    | 17    | 17    |
| All-Sky Solar Radiation                                           | RadAvg      | 344    | 662   | 1110  | 1541  | 1729  | 1868  | 1860  | 1590  | 1132  | 663   | 331   | 248   | 248   |
|                                                                   | RadStd      | 43     | 64    | 99    | 142   | 122   | 129   | 137   | 128   | 80    | 81    | 36    | 30    | 30    |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page